

# Module 1 Debugging Exercise

C++ fundamentals for VEX

Student: \_\_\_\_\_

Date: \_\_\_\_\_

Pathway: Code Learning / Robot Programming

Team: \_\_\_\_\_

## Part A: Repair the Code

Find and correct every error. Explain each correction beside the code.

### Exercise 1

```
int speed = 6
speed += 2;
```

### Exercise 2

```
bool pistonState = 4.5;
```

### Exercise 3

```
if (motorPort = 3) {
  motorPort = 4;
}
```

### Exercise 4

```
double distance = measurDistance();
```

### Exercise 5

```
// movement.cpp
void testMove() { move(50, 0, 6); }

// auto.h
// Add the missing declaration here.
```

## Part B: Predict the Result

```
int speed = 5;
speed += 2;

if (speed > 6) {
    speed = 6;
}
```

Final value of speed: \_\_\_\_\_ Explain: \_\_\_\_\_

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## Part C: Write a Function

```
double limitVoltage(double requestedVoltage, double maximumVoltage);
```

Write the definition. It must limit both positive and negative requested voltage.

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## Declaration vs. Definition

In your own words, explain the difference and name the file where each normally belongs.

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## Checkpoint

Module quiz score: \_\_\_\_\_% Minimum passing score: 75%